Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 04
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. If $a = 2^2 \times 3^3 \times 5^4$ and $b = 2^3 \times 3^2 \times 5$, then HCF$(a, b)$ is equal to: [1]
(a) 90 (b) 180 (c) 360 (d) 540

Q2. The total number of factors of a prime number is: [1]
(a) 1 (b) 2 (c) 0 (d) 3

Q3. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is: [1]
(a) 4 (b) 2 (c) 1 (d) 3

Q4. If p and q are two co-prime natural numbers, then HCF(p, q) is equal to: [1]
(a) pq (b) $p + q$ (c) 1 (d) 0

Q5. **Assertion (A):** $\sqrt{2}$ is an irrational number.
Reason (R): If p is a prime number, then $\sqrt{p}$ is an irrational number. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Given that HCF$(306, 657) = 9$, find LCM$(306, 657)$. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Find the HCF and LCM of 288, 360, and 384 by the prime factorisation method. [3]

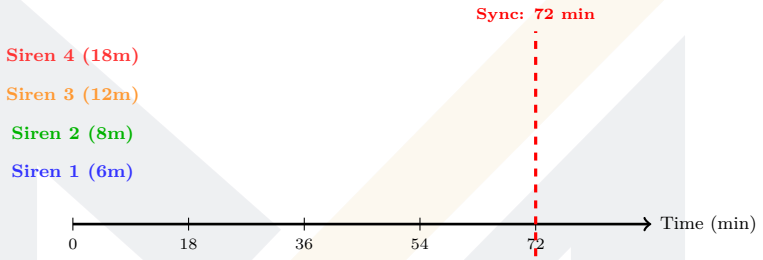
Q8. An army contingent of 616 members is to march behind an army band of 32 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march? [3]

Q9. A sweet seller has 420 kaju burfis and 130 badam burfis. She wants to stack them in such a way that each stack has the same number, and they take up the least area of the tray. What is the number of burfis that can be placed in each stack for this purpose? [3]

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: CASE STUDY 04: INDUSTRIAL ELECTRONIC ALARM CHIMES WARNING SIRENS [4 Marks]

A modern automated production factory employs a synchronized auditory signal system to coordinate manufacturing shifts, machine safety checks, cooling inspections, and supervisor walk-throughs. Four distinct sirens sound automatically at regular intervals of 6 minutes, 8 minutes, 12 minutes, and 18 minutes respectively. All four sirens are synchronized and sound simultaneously at 8:00 AM.



(i) Find the least common multiple (LCM) of 6, 8, 12, and 18 using the prime factorisation method. [2]

(ii) At what exact clock time will all four sirens next sound simultaneously? [1]

(iii) How many times will all four sirens sound simultaneously during a 12-hour factory shift (including the initial toll at 8:00 AM)? [1]

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